

Atomic and topological structure of domain wall networks in commensurate charge density wave of $2H\text{-TaSe}_2$

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Competing periodicities and the symmetry breaking in quantum materials often produce domain wall networks (DWNs) of localized topological defects with macroscopic connectivity. However, because such defects are extremely dilute in commensurate charge-density-wave (CDW) phases, the intrinsic structures and topology of DWNs in this regime have been elusive. Here, using large-scale scanning tunneling microscopy measurements, we unveil a DWN embedded in the prototypical commensurate CDW phase of $2H\text{-TaSe}_2$. We unambiguously determine the atomic structures of topological defects, and perform a systematic topological classification of their connectivity. This analysis establishes a honeycomb network built from a unique $2\pi/3$ domain wall structure and two different Z_3 vortices. This DWN is qualitatively distinct from those reported in incommensurate CDW phases, yet realizes the canonical Z_3 honeycomb network predicted for hexagonal systems with competing periodicities. These findings uncover the hidden landscape of commensurate CDW domains and provide a generic tool for understanding topological structures of Z_3 DWNs in a broad class of physical systems including CDWs, moire superlattices, and ferroic materials.

Introduction - Competing periodicities in lattice, magnetic [1, 2], charge [3–5], or polarization [6–10] orders often produce domain wall networks (DWNs), which are macroscopic patterns built from spatially confined topological defects such as domain walls (DWs) and their vortices [11]. As recently demonstrated in twisted stacks of van der Waals materials, competing or incommensurate lattice periodicities generate networks of localized dislocation DWs [12–18], whose local structures and/or properties differ substantially from those of the surrounding ground state. These defects are endowed with topological charges which cannot be altered without destroying the host order parameters. Such distinct properties and inherent robustness make them powerful tools for engineering physical properties and novel functionalities of host materials [2, 19]. Beyond individual defects or simple defect assemblies, the DWN as a whole can accommodate emergent collective properties [20, 21] and its topological property serves as a direct readout of the underlying degeneracy of the system [11].

A charge density wave (CDW), a periodic modulation of charge density competing and interacting with the lattice periodicity [22], is a paradigmatic setting in which topological defects and their networks play important roles. Upon cooling, a CDW system typically undergoes phase transitions from a normal unmodulated state to an incommensurate CDW (ICDW) phase, where the CDW and the host lattice have incommensurate periodicities, and eventually to a commensurate CDW (CCDW) phase [23]. DWs and their vortices naturally arise in the ICDW phase to form DWN and evolve across the ICDW-CCDW transition [24]. These DWNs have been related to emerging quantum phases such as superconductivity emerging from suppressed CDW states [21].

Recent progress in scanning tunneling microscopy

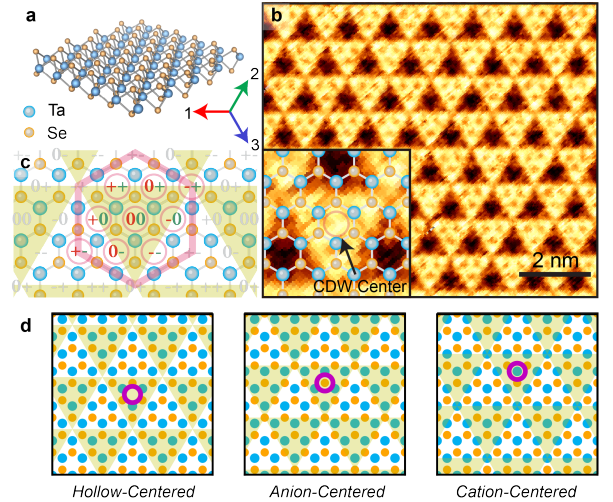


FIG. 1. (a) Schematics of the single-layer atomic structure of a $2H\text{-TaSe}_2$ crystal. (b) High-resolution filled-state (-400 mV sample bias) STM topography image of the 3×3 CDW phase at 78 K. The inset shows the correspondence of the topography to the atomic structure. (c) Schematic representation of the CDW unit cell (the red hexagon or a white and a yellow triangle). A CDW maximum with its center at the (0,0) hollow site can be translated to eight other hollow sites of the given unit cell indicated by red circles with corresponding phase shifts to constitute nine degenerate CDW configurations (see the main text for the phase shift notations). The above arrows indicate three unit vectors, where the third one is redundant. (d) Schematic representations of three high-symmetry CDW structures with respect to the atomic lattice; the CDW maximum (magenta circle) is centered at a hollow, a cation (sky-blue ball), and an anion site (orange ball), respectively.

(STM) has enabled atomic-scale observation and charac-

terization of DWs and DWNs in several model CDW systems. For example, two different DW structures with in-gap electronic states [4] were identified in the $\sqrt{13} \times \sqrt{13}$ CCDW of 1T-TaSe₂. In the higher-temperature ICDW phase, a honeycomb DWN with Z_3 vortices was found to form with a different type of DWs [25]. In the 2×2 CCDW phase of 1T-TiSe₂, a unique DW connecting domains of different chirality without any in-gap electronic state was identified [26–28], yet the associated DWN has not been identified. In 2H-NbSe₂ with a 3×3 CDW, STM studies revealed a distinct DW structure connecting two energetically degenerate CDW structures, called split DWs, which forms an entangled CDW ground state. The nature DWN in this ICDW phase remains under debate between a honeycomb structure with Z_3 vortices [3] and a triangular structure with Z_6 vortices [3, 5]. This type of a ‘split’ DW were originally predicted for a twin system of 2H-TaSe₂ [29], whose DWs have not been investigated in detail.

Despite these intensive efforts, a well-characterized DWN in the CCDW phase has remained elusive even in these canonical systems. The main obstacle are the very low density of DWs for STM investigations of the whole DWN and the remarkable richness of structural and electronic degrees of freedom within CDW DWs as revealed in recent works. While there have been earlier electron microscopy studies for bulk crystals, these studies often did not provide atomistic details and have limitations in addressing 2D DWNs within individual layers. Although the classical McMillan theory of the ICDW-CCDW phase transition [24] describes the evolution of DWNs and DWs, including the conservation of the network’s topological properties and the narrowing of DWs, explicit experimental verification of these theoretical predictions has not been achieved and contradicting observations have reported as noted above. Given the intricate DW structures uncovered in recent studies, the experimental characterization of the converged DW structures and DWN topology in a CCDW phase thus represents an outstanding and timely scientific problem.

In the present work, we identified and characterized a DWN of a prototypical CCDW phase found in 2H-TaSe₂. While previous electron microscopy studies revealed the existence of DWs and orthorhombic domain structures in this phase [30–32], atomistic details were lacking and the topological nature of the 2D DWN was elusive. In sharp contrast, we observed a honeycomb network and revealed the atomic structures of the DWs and DW vortices involved. The topological nature of the DWN was analyzed, which is consistent with the $Z_3 \times Z_3$ symmetry of the CDW. We demonstrate the conservation of the topological charges within this DWN through the boundary-charge correspondence of the network, thereby establishing the topological nature of the DWN, and generalize this approach to various other DWNs.

2H-TaSe₂ CDW structure & degeneracy - 2H-

TaSe₂ crystallizes in a hexagonal lattice (space group $P6_3/mmc$), characterized by a stacking of two Se-Ta-Se layers with each Ta atom coordinated by six Se atoms in a trigonal geometry [33]. As the temperature decreases below 118 K, the system develops an ICDW state, which transits to a 3×3 CCDW phase below 87 K ($Cmcm$) [34]. The STM topography obtained at -400 mV in Fig. 1(b) shows the well-known topography of the 3×3 CDW structure with an alternating triangular pattern with six (bright) and three (dim) protrusions. The atomic lattice overlaid in the inset shows that the CDW center (center of a bright triangle) is located at a hollow site surrounded by Ta and Se atoms. Previous theoretical studies suggested three low-energy CDW structures: hollow-centered (HC), cation(Ta)-centered, anion(Se)-centered, according to the location of CDW maxima [Fig. 1(d)] [35] and our STM image matches the HC structure in agreement with previous STM studies [36–38]. While this structure was found to coexist with the anion(Se)-centered structure in the sister material 2H-NbSe₂ [3, 5, 39], our survey over very large areas confirmed a unique CDW structure. This can be understood from the fact that the energy difference between the HC and anion(Se)-centered structures is two-times larger in 2H-TaSe₂ than in 2H-NbSe₂ [3, 40].

The translational symmetry breaking into the 3×3 superlattice leads to nine degenerate ground states with $Z_3 \times Z_3$ translational order; three translational degeneracy in three-rotationally degenerate CDW wave vectors [$3\mathbf{a}_1$, $3\mathbf{a}_2$, and $3\mathbf{a}_3$ in red, green, and blue arrows, respectively, in Fig. 1(a)]. The translation of a 3×3 CDW unit cell can be defined by the lattice vector $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2$. The relative phase of an individual domain is uniquely determined by the coefficients $(n_1, n_2) \pmod{3}$ (the values $\{0, 1, 2\}$ are represented by $\{0, +, -\}$ in the following), such as $(0, +)$, $(-, -)$, or $(+, -)$ as shown in Fig. 1(c).

CDW domain wall network - Figure 2(a) shows an enlarged STM image for a DW connecting two degenerate CDW domains observed at 78 K. We find that the CDW periodicity is frustrated at the DW region with a CDW phase shift of a single lattice unit vector ($2\pi/3$). We then measured CDW phase maps by applying the spatial lock-in method to large-scale STM topographies [Figs. S1-S4 in Supplementary Materials] [2, 5, 41] to characterize the distribution of DWs in large areas [Fig. 2(e)]. We identified DWs with six different phase shifts of $\pm(+, 0)$, $\pm(0, +)$, and $\pm(-, -)$ [Figs. 2(a), 2(b), and 2(c), respectively], which correspond to the basis vectors $\pm\mathbf{a}_1$, $\pm\mathbf{a}_2$, and $\pm\mathbf{a}_3$, respectively. The three DWs tend to align perpendicularly to the corresponding basis vector. All DWs observed have the same atomic structure shown in Fig. 2(a), which has a very narrow width of about $7a$. In the DWN shown in Fig. 2(e), we can find that the shortest path connecting the same domains [two $(+, 0)$ or $(+, -)$ domains] crosses three DWs with identical phase shifts. This clearly demonstrates the transla-

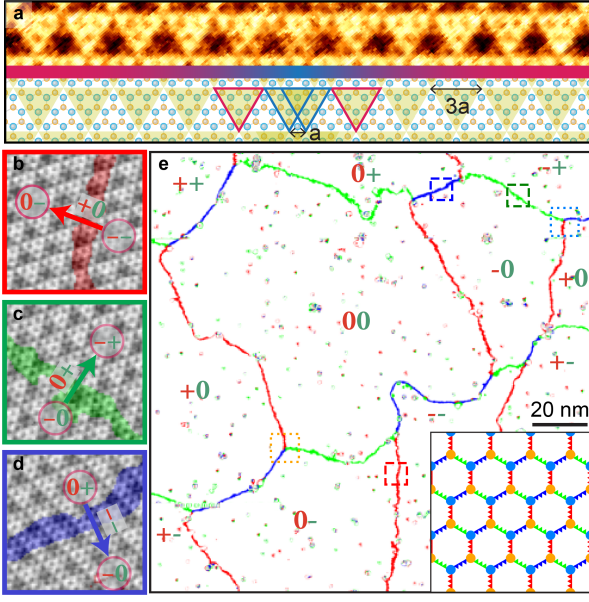


FIG. 2. (a) Enlarge STM image crossing a phase shift domain wall. The corresponding atomic and CDW structures are shown at the bottom, indicating that the phase shift corresponds to a single primitive unit cell (a). In between the top and the bottom panel, the phase map along the centers of the CDW maxima is plotted: red (blue) color presents the CDW maxima on a hollow site. (b–d) STM topography images with numerically-extracted phase-shift maps overlaid (represented by the color density) for three DWs with phase shifts of the same size but three different directions. These images are enlarged from a large scale image shown in e for the boxed areas of corresponding colors. The phase shifts, $+0$, $0+$, and $-$, are indicated, connecting domains specified by the corresponding indices with red circles. (e) A large-area phase shift map extracted from an STM image, revealing an extended network of the CDW domain walls. The red, green, and blue lines correspond to the DWs characterized in b, c, and d, respectively. The inset shows a schematic model of the honeycomb DWN, where three different DWs are given with their directions specified by the arrow heads and their two different vortices by dots of different colors. Point defects with finite local CDW phase shifts are also observed, which are not discussed here.

tional Z_3 symmetry. Note that DWs involving $\pm(+, -)$ shifts were not observed, due probably to a higher energy cost associated with a larger phase shift, in consistency with the theoretical prediction [24, 42].

Z_3 vortices - We also find that the DWs are all connected by Z_3 vortices, where three different types of DWs merge to form a honeycomb DWN depicted ideally in the inset of Fig. 2(e). The vortices can be classified into two types with the phase shifts surrounding them as shown in Figs. 3(a) and 3(b). We define a phase shift loop around a vortex by sequentially recording the phase shifts along a clockwise loop encircling the vortex cores. The phase shift loop is rgb or $r^{-1}g^{-1}b^{-1}$ for two different types of vortices observed, where r, g, b represent the

phase shifts $(+, 0)$, $(0, +)$, and $(-, -)$, respectively. The high-resolution STM topography [Figs. 3(f) and 3(g)] further reveals that the vortex core has a distinct local CDW registry from domains and DWs. In an rgb vortex, the high-symmetry points of charge density are located on the Ta sites, similar to the local cation-centered CDW in Fig. 3(d). In sharp contrast, the $r^{-1}g^{-1}b^{-1}$ vortex is centered on the Se site, similar to the anion-centered CDW. In fact, the phase shift loop dictates the above atomic and CDW registry of a vortex as shown in the insets of Figs. 3(c) and 3(d). Namely, the rgb ($r^{-1}g^{-1}b^{-1}$) phase shifts must involve a rotation centered on a Ta (Se) atom.

The observed DWN is qualitatively different from the previous STM findings in comparable $2H$ -polytype structures. In ICCDW phase of $2H$ -NbSe₂, the ground state involves the coexistence of both HC and AC structures [3, 5]. The competing structure makes the DWs split and connect different CDW structures as theoretically predicted [29], which, however, does not apply to the present system with a unique HC ground state. Furthermore, the disordered DWN have prevented the clear identification of vortex structures, leading to conflicting DWN models with Z_3 [3] or Z_6 [5] vortices. On the other hand, while Pd-intercalated $2H$ -TaSe₂ in the ICDW state shares a single HC ground state with our findings, it exhibits a DWN with Z_6 vortices involving the breaking of both translational and rotational symmetries [38], in stark contrast to the present observation. However, it is notable that the present DWN structure is very similar to that observed in the ICDW (also called nearly commensurate CDW) phase of $1T$ -TaS₂ [25] with a much larger degeneracy. Note that the width of the DWs (domains) in the present case is much narrower (wider) than the ICDW phase of $1T$ -TaS₂, which is consistent with the prediction of the ICDW-CCDW transition in the MacMillan theory [24].

Early electron microscopy (EM) studies on $2H$ -TaSe₂ [31, 32] observed the formation of orthorhombic domains and DWNs in bulk samples, which are distinct from the present observations regarding both the density of DWs and the symmetry and topology of the DWNs. We note that EM did not resolve individual domains within a single layer because it primarily captures changes in the stacking order. As the previous theories suggested [24, 43, 44], the paired domain walls between neighboring layers do not disturb the stacking order and cannot be detected in EM. This may explain the difference from the present STM observation. These EM works also suggested the DW splitting due to competing CDW structures as the origin of the conversion from a honeycomb to an orthorhombic DWN. We however do not find any competing CDW structures for the DW splitting as noted above.

Topological nature of the domain wall network

- The topological nature of the network can be viewed

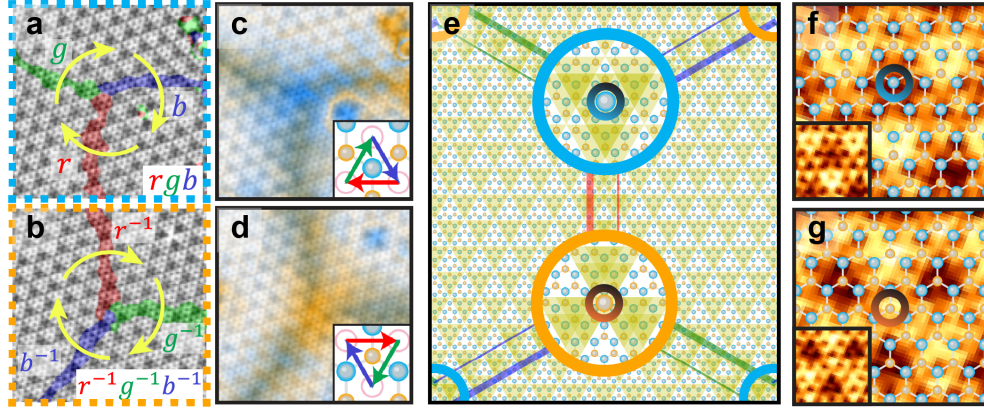


FIG. 3. (a, b) STM topography images and phase-shift maps of two distinct DW vortices, which connect three different types of DWs (red, green, and blue) shown in Fig. 2 in clockwise order (denoted as rgb and $r^{-1}g^{-1}b^{-1}$, respectively). (c, d) The same STM topography images with (a, b) overlaid with the phase shift components of CDW maxima on a Ta atom and a Se atom in sky-blue and orange colors, respectively. In these maps, the locations of DWs are highlighted by the sky-blue contrast (CDW maxima in Ta sites). The locations of two vortices are distinct. The insets show the trajectory of the CDW phase crossing three DWs involved in the order parameter space. (e) Schematic model of the basic elements of the DWN observed with two vortices, whose structures are enlarged within the circles. (f, g) Atomic-resolution STM images with the underlying atomic structures for rgb and $r^{-1}g^{-1}b^{-1}$ vortex cores, respectively.

from the boundary-charge correspondence [2, 13]. Since each DW has a phase shift, it is endowed with a fractional charge as in the case of a soliton. When an arbitrary closed boundary is given, one can define the total amount of the boundary charges for all DWs crossed by the boundary. This surface charge is topologically connected to the total integrated topological charges of all vortices within the boundary [2, 13]. The boundary charge, $\Phi(L)$, is defined by the sequence of phase shifts across the intersected DWs by a closed path (e.g., $\Phi(L_A) = rrg^{-1}b$). We introduce topological charges q_C and q_A for the rgb and $r^{-1}g^{-1}b^{-1}$ vortices, respectively, and define the bulk topological charge as $Q(L) = q_C^{n_C} q_A^{n_A}$, where n_C (n_A) is the number of rgb ($r^{-1}g^{-1}b^{-1}$) vortices enclosed. The boundary-charge correspondence states that $\Phi(L)$ uniquely determines $Q(L)$, and this can be demonstrated geometrically and algebraically.

In the geometric approach, $\Phi(L)$ corresponds to a closed path in the order parameter space introduced in Fig. 1(c). This path encloses a specific area, which can be triangulated in the way shown in Figs. 3(c) and 3(d). The number of enclosed triangles directly determines the net topological charge: the number of upward (downward) triangles corresponds to n_C (n_A) [Fig. 4(b)]. The cases for L_A and L_B are given in Fig. 4(c). In the algebraic approach, the application of group multiplication rules reduces $\Phi(L)$ into the form of $q_C^{n_C} q_A^{n_A}$. Alternatively, we can determine the topological charge algebraically by treating the boundary sequence as a product of non-Abelian group elements. By applying the group operation rules—where a vortex topological charge is defined as $q_C = rgb$ ($r^{-1}g^{-1}b^{-1}$) and that of a vortex pair is denoted by the commutator $q_C q_A = r^{-1}g^{-1}rg = [r, g]$ —an

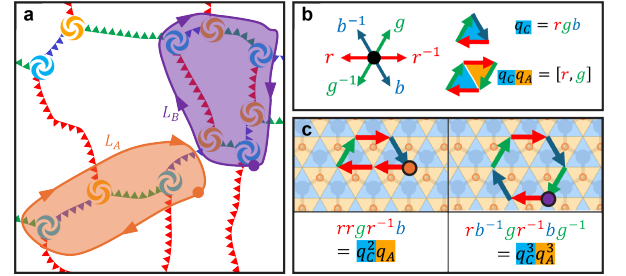


FIG. 4. (a) Schematic illustration of the DWN observed in Fig. 2(e) with different DW and vortex types specified with symbols as introduced in Fig. 3. Two representative closed loops, L_A and L_B , enclose different configurations of topological defects. (b) (Left) rgb ($r^{-1}g^{-1}b^{-1}$) phase shift vectors in the order parameter space. (Right) The phase shift loops for a single rgb vortex and two connected vortices of rgb and $r^{-1}g^{-1}b^{-1}$, which are visualized geometrically and expressed algebraically. q_C and q_A represent the topological charges of rgb and $r^{-1}g^{-1}b^{-1}$ vortices, respectively. (c) Bulk topological charge calculations for the loops L_A and L_B via geometric (top) and algebraic (bottom) approaches.

arbitrary boundary sequence $\Phi(L)$, such as $rrgr^{-1}b$, can be systematically reduced to $q_C^2 q_A$. To provide a more practical framework for the algebraic reduction in any arbitrary closed boundary, we can map the phase shifts and the topological charges onto the Heisenberg group $\mathcal{H}$ as shown in the ?? Supplementary Material.

In addition to a honeycomb DWN with Z_3 vortices, the incommensuration in a triangular lattice can also result in an alternating triangular DWN with Z_6 vortices as discussed in a twisted graphene moiré lattice [18]. A later work showed the boundary-charge correspondence

for this type of DWN [13]. Our topological framework is consistent with this work and is applicable to DWNs featuring Z_3 or Z_6 vortices arising from various competing periodic systems, such as transition metal dicalcogenides [3, 5, 25], marginally twisted bilayers [13, 15, 18] and ferroic hexagonal manganites [6–10] (see Fig. S7 in the Supplemental Material).

Summary - We have elucidated the intrinsic DWN in the commensurate CDW phase of $2H$ -TaSe₂ by combining large-scale atomic-resolution STM imaging and topological classification. We identified the topological defects, namely a unique DW structure with a $2\pi/3$ phase shift and two Z_3 vortices, which organize into a honeycomb DWN with a length scale of 60 nm. While the DWN characterization has not been available for commensurate CDW systems, the present DWN is distinct from those observed in the incommensurate CDW phases of $2H$ -NbSe₂ and Pd-intercalated $2H$ -TaSe₂. We further establish the boundary-charge correspondence of the topological charges in a Z_3 honeycomb DWN. Our work experimentally realizes a long-predicted ideal DWN in a CDW phase with hexagonal symmetry and provides a generic method to understand the topological properties of the networks emerging in systems with competing periodicities ranging from transition-metal dichalcogenides to moiré heterostructures and hexagonal ferroics.

Method

Commercial single crystals of $2H$ -TaSe₂ (HQ graphene) were cleaved in ultrahigh vacuum at room temperature to obtain clean surfaces and transferred to the ultrahigh-vacuum chamber equipped with a commercial cryogenic STM operating at 77 K. The STM tip was fabricated by cutting Pt-Ir wire. All STM images were obtained at -400 mV using the constant current mode.

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